

SHERBAKUL', Russian Federation

WMO: 287910

Lat: 54.633N

Lon: 72.433E

Elev: 377

StdP: 14.50

Time Zone: 6.00 (E06)

Period: 94-18

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
1	-29.9	-24.5	-34.9	0.8	-29.6	-29.5	1.1	-24.2	23.4	16.3	21.1	19.7	3.7	220	0.644

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	18.2	87.1	65.3	83.7	64.5	80.7	63.8	69.0	80.6	67.3	78.6	65.8	76.5	9.1	220

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	
65.1	94.1	73.5	63.4	88.7	72.1	61.7	83.4	70.6	33.4	80.1	32.1	78.6	30.8	76.5	74.1

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	DB	WB
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(4) 22.3	19.4	17.5	-33.3	93.1	7.5	3.1	-38.7	95.3	-43.0	97.1	-47.2	98.8	-52.7	101.1		
(5)			-33.7	71.6	7.0	1.5	-38.7	72.6	-42.8	73.5	-46.7	74.3	-51.8	75.4	(4)	(5)

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec		
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(6) Temperatures, Degree-Days and Degree-Hours	DBAvg	35.4	-1.1	3.1	18.1	39.4	54.7	64.8	66.0	62.1	51.4	38.1	20.1	5.7		
	DBStd	26.30	14.35	12.96	11.19	10.65	9.00	7.71	6.56	6.93	8.00	8.60	13.29	14.35		
	HDD50	7014	1583	1312	989	344	57	3	0	3	76	377	897	1373		
	HDD65	11079	2048	1732	1454	769	335	98	71	138	414	835	1347	1838		
	CDD50	1675	0	0	0	25	202	448	497	379	118	6	0	0		
	CDD65	266	0	0	0	0	14	93	103	49	7	0	0	0		
	CDH74	2826	0	0	0	10	274	995	921	507	119	0	0	0		
	CDH80	806	0	0	0	1	58	325	259	136	27	0	0	0		
	WSAvg	7.9	7.5	7.9	9.1	9.0	8.8	7.3	6.8	6.7	7.2	7.8	8.5	8.4		
	PrecAvg	14.2	0.6	0.6	0.6	0.9	1.2	1.7	2.5	2.2	0.9	1.1	1.1	0.8		
	PrecMax	20.1	1.5	1.5	1.7	2.0	4.8	3.4	4.9	4.8	2.0	2.8	2.8	1.6		
(15) Precipitation	PrecMin	8.5	0.0	0.0	0.1	0.2	0.1	0.5	0.5	0.7	0.3	0.1	0.2	0.2		
	PrecStd	2.9	0.4	0.3	0.4	0.5	0.9	0.6	1.3	1.0	0.5	0.7	0.6	0.4		
	DB	32.4	33.7	44.1	74.2	87.1	91.1	91.5	89.1	83.4	65.6	44.8	34.5			
	MCWB	31.4	31.9	39.2	55.0	61.2	64.7	67.5	65.8	60.5	51.7	41.9	32.4			
	2% DB	28.4	30.5	37.3	67.2	81.3	87.3	85.9	83.5	77.2	61.2	41.3	32.3			
(23) Mean Coincident Wet Bulb Temperatures	MCWB	27.2	29.3	34.5	51.0	58.4	65.1	66.4	64.9	57.6	49.5	39.3	31.0			
	DB	25.0	27.3	35.4	62.6	77.0	83.9	82.5	79.5	71.0	56.0	38.4	29.1			
	MCWB	24.1	26.2	33.3	49.6	57.3	64.7	66.1	64.1	56.7	47.1	36.1	27.8			
	10% DB	21.0	23.2	33.4	57.0	72.4	79.8	79.2	75.9	66.0	51.6	35.0	24.8			
	MCWB	20.0	22.1	31.6	47.3	55.3	63.6	65.2	62.6	54.0	45.1	33.2	23.4			
(27) Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4% WB	31.5	32.3	39.3	56.8	63.2	70.9	70.9	69.8	63.6	54.3	42.9	33.0			
	MCDB	32.5	33.5	43.4	70.6	80.2	83.9	81.2	80.5	78.5	63.0	44.1	33.7			
	2% WB	27.2	29.3	35.0	53.1	60.6	68.4	69.3	67.4	60.2	50.9	39.5	31.1			
	MCDB	28.1	30.4	36.9	64.0	77.3	81.5	80.5	77.7	72.4	58.2	41.4	32.3			
	5% WB	24.1	26.2	33.5	50.2	58.5	66.4	67.8	65.7	57.5	48.1	36.1	28.0			
(33) Mean Daily Temperature Range	MCDB	25.0	27.3	35.2	61.2	72.9	78.4	79.1	76.2	68.5	54.9	38.0	29.2			
	10% WB	20.1	22.1	31.7	47.6	56.6	64.5	66.4	64.0	54.9	45.4	33.4	23.5			
	MCDB	21.1	23.2	33.4	57.0	69.9	77.5	76.7	73.7	64.4	51.2	34.8	24.7			
	5% DB	13.8	15.0	15.3	17.8	22.0	20.3	18.2	18.9	19.1	14.7	10.9	13.5			
	MCDBR	13.9	12.6	12.0	28.3	29.8	26.8	24.1	26.5	28.2	24.6	11.3	13.4			
(40) Clear-Sky Solar Irradiance	MCWBR	13.7	12.2	10.4	14.8	12.3	10.2	9.1	11.3	12.8	14.3	9.9	13.1			
	5% WB	13.6	12.6	12.1	25.7	25.5	22.6	20.2	21.3	24.4	21.0	10.7	13.6			
	MCDBR	13.6	12.4	10.8	14.5	12.4	10.1	8.9	10.1	12.1	13.3	9.7	13.3			
	taub	0.286	0.317	0.375	0.408	0.404	0.409	0.435	0.432	0.371	0.353	0.324	0.290			
	taud	2.048	2.018	1.943	2.129	2.223	2.293	2.207	2.183	2.313	2.242	2.108	2.014			
(42) All-Sky Solar Radiation	Ebn at Noon	196	233	244	257	266	266	255	247	249	221	178	161			
	Edh at Noon	22	33	45	43	41	39	42	40	31	26	21	18			
	RadAvg	324	670	1164	1479	1807	1919	1760	1435	1029	534	255	209			
(45)	RadStd	23	28	98	121	154	103	130	109	78	69	36	24			

Historical Trends

		DBAvg	Heating			Cooling			Degree-Days			
			99% DB	99% DP	1% DB	1% WB	1% DP		HDD50	HDD65	CDD50	CDD65
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	
(46)	Station Only	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
(47)	Regional (9 neighbors)	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A

Nomenclature: See separate page